Tectonophysics, 216 (1992) 365–378
Elsevier Science Publishers B.V., Amsterdam

Paleomagnetic data from Umbria (Italy): implications for the rotation of Adria and Mesozoic apparent polar wander paths

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(Received December 17, 1991; revised version accepted May 20, 1992)

ABSTRACT

Channell, J.E.T., 1992. Paleomagnetic data from Umbria (Italy): implications for the rotation of Adria and Mesozoic apparent polar wander paths. *Tectonophysics*, 216: 365–378.

Sixteen new Jurassic and Cretaceous paleomagnetic sampling sites are located in northern Umbria along three river valleys (Bosso, Burano, Sentino). The Late Jurassic and Early Cretaceous data are consistent in declination, but not inclination, with previously published data from these valleys. Early Jurassic data from eight sites give site mean declinations which are significantly different from the two site mean declinations of this age which were previously available. Comparison of the data with Cretaceous and Jurassic mean directions from the Southern Alps indicate 20° anticlockwise rotation, relative to the Southern Alps, of this part of northern Umbria during Neogene deformation of the Umbrian fold and thrust belt. If the Umbrian mean directions are rotated to compensate for the tectonic rotation, the Early Jurassic and Cretaceous Umbrian and Southern Alpine pole positions coincide closely with a recently derived African apparent polar path (APWP), implying that Adria moved with Africa since at least Early Jurassic times. The *Late Jurassic* Umbrian and Southern Alpine poles, however, are not consistent with current APWPs and imply rapid apparent polar wander (in both African and North American coordinates) from high latitude to lower latitude between Bathonian and Oxfordian times.

Introduction

The Umbrian (or Umbrian–Marches) Apennines are an east- and northeast-vergent arcuate-shaped fold and thrust belt in central Italy (Lavecchia et al., 1988). In northern Umbria (Fig. 1), three rivers cut through NW–SE-oriented anticlines, providing excellent exposure of Umbrian Jurassic and Cretaceous stratigraphy. These are classic sections in southern Tethyan Mesozoic stratigraphy, with Rhaetian–Hettanginian shallow-water Calcare Massiccio limestones succeeded by Sinemurian–Eocene pelagic and seamount facies calcareous and siliceous sediments, recording the Early Jurassic foundering of the carbonate platform and subsequent basinal development (Bortolotti et al., 1970; Bernoulli and Jenkyns, 1974; Cresta et al., 1989). The principal biostratigraphic control in the basinal sediments is from foraminifera in the Upper Cretaceous Scaglia, calcareous nannofossils in the Lower Cretaceous Maiolica, ammonites in the Toarcian–Aalenian Ammonitico Rosso and calcareous nannofossils in the Pliensbachian Corniola (see Cresta et al., 1989).

The Umbrian Apennines have been the focus for magnetostratigraphic studies of Jurassic to Miocene pelagic sediments, such as those at Gubbio (e.g., Lowrie and Alvarez, 1977; Lowrie et al., 1982; Napoleone et al., 1983), at Fonte del Giordano (Fig. 1; Cirilli et al., 1984), at Presale (Fig. 1) (Lowrie and Alvarez, 1984), at Bosso (Fig. 1; Lowrie and Channell, 1984) and at Sentino (Fig. 1; Channell et al., 1984). The results from these and other magnetostratigraphic studies in the region have played an important role in establishing the correlation of microfaunal biozonations to the magnetic polarity timescale.

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A number of studies have focused on tectonic interpretation of Umbrian paleomagnetic data. Lowrie and Alvarez (1975) and Channell and Tarling (1975) noted that Cretaceous paleomagnetic directions from northern Umbria are rotated anticlockwise relative to coeval directions from the Southern Alps and Africa. VandenBerg et al. (1978) showed that the Jurassic–Cretaceous paleomagnetic directions have a pattern similar to that expected for Africa, but are rotated 25° anticlockwise. The anticlockwise rotation was considered to be a post-Early Eocene rotation of the entire Italian peninsula, as advocated earlier by Lowrie and Alvarez (1975). An alternative interpretation is that the rotation is restricted to the sedimentary cover and is a result of deformation in the Umbrian fold and thrust belt (Channell and Tarling, 1975). This explanation is supported by an analysis of paleomagnetic data from throughout Umbria which indicated variation in paleomagnetic declination, with more westerly declinations in northern Umbria (Channell et al., 1978). An apparently progressive change in fold axis orientation from approximately north–south in central/southern Umbria to northwest–southeast in northern Umbria was interpreted to reflect differential amounts of eastward transport of the sedimentary cover and anticlockwise rotation of the northern Umbrian sedimentary cover. A recent study of the Lower Cretaceous Maiolica from throughout Umbria (Hirt and Lowrie, 1988) indicates that, although the paleomagnetic *inclinations* conform to a Fisherian distribution, the declinations do not. The declination variability is particularly apparent in central/southern Umbria, and is clearly the result of rotations associated with deformation of the sedimentary cover. Hirt and Lowrie (1988) confirmed that declinations from northern Umbria are more westerly than those from central/southern Umbria and, by plotting declination against fold axis orientation, demonstrated that paleomagnetic declinations are not simply related to fold axis orientation. This result tends to indicate that the variation in fold axis orientation does not postdate the formation of the folds, indicating that Umbrian Apennines are not, strictly speaking, an orocline. The deformation style in the fold and thrust belt must involve pre- and synfolding rotation.

Márton and D'Andrea (1992) analysed six sites from Sinemurian to Santonian sediments at Presale (Fig. 1). They compared the Cretaceous paleomagnetic data with those from Gargano (southern Italy) and concluded that the Presale area rotated 20–25° anticlockwise relative to Gargano in Tertiary times. By comparing the data from

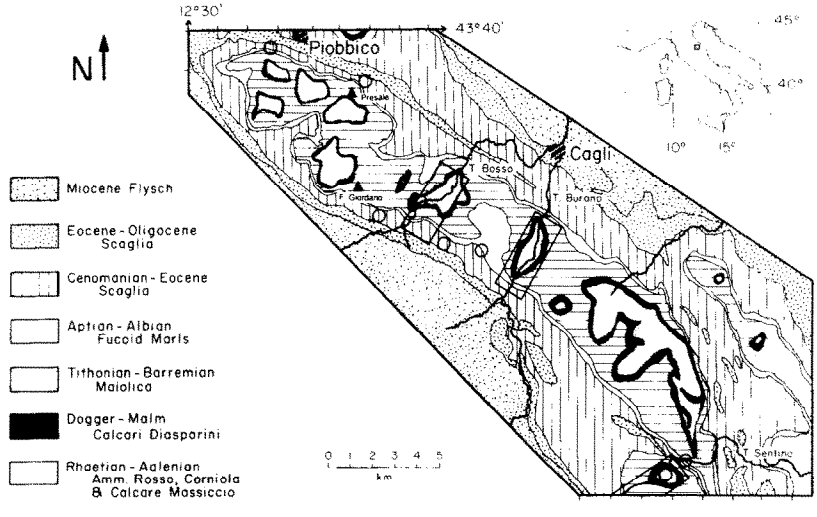


Fig. 1. Generalized geologic map of northern Umbria (after Carta Geologica d'Italia, 1:100,000, Sheet 116, Gubbio; Bortolotti et al., 1970; VandenBerg et al., 1978). Circles mark the sampling sites of VandenBerg et al. (1978). Triangles mark the Presale sampling site of Lowrie and Alvarez (1984) and Márton and D'Andrea (1992), and the Fonte del Giordano sampling site of Cirilli et al. (1984). The sections sampled in this study (Bosso, Burano, Sentino) are boxed.

Gargano with the Cretaceous African data of Hargraves (1989), Márton and D'Andrea (1992) concluded that Gargano was rotated 11° relative to Africa, implying a 31–36° anticlockwise rotation of the Presale area relative to Africa.

Paleomagnetic sampling

The objective of this study was to re-evaluate the "African" character of Umbrian paleomagnetic data, which has been based largely on the paleomagnetic data of VandenBerg et al. (1978). Apart from this important study, the vast majority of the available magnetostratigraphic and paleomagnetic data in Umbria are from Cretaceous rocks, with few Jurassic data. In order to compare the pattern of Umbrian paleomagnetic directions with the African pattern, it is necessary to supplement existing Jurassic data.

The deformation-related variation in paleomagnetic declination in the Umbrian Apennines requires that we restrict our sampling to individual sections which expose the Jurassic–Cretaceous stratigraphy, in order to minimize the likelihood of tectonic rotation between sites. We chose three NE–SW-oriented river valleys, separated by up to 20 km, which cut the Umbrian Apennines in northern Umbria (Fig. 1). These valleys (Burano, Bosso, Sentino) expose the Liassic to Upper Cretaceous stratigraphy in single sections. This is the same region as that sampled by VandenBerg et al. (1978), although their sampling extended a further 20 km to the north (Fig. 1). Our northernmost sampling locality (Bosso) is 7 km southeast of the Presale sampling area of Márton and D'Andrea (1992) (Fig. 1).

Sixteen new sites were collected from the Bosso, Burano and Sentino river valleys (Fig. 1; Table 1). The sampled formations are the Upper Cretaceous Scaglia, Lower Cretaceous Maiolica, Kimmeridgian–Tithonian Calcari Diasparini, Toarcian–Aalenian Ammonitico Rosso and Pliensbachian Corniola. For individual sites, samples were collected over 2–10 m of stratigraphic section. The data were supplemented by four previously published site mean directions from the same valleys, two from Hirt and Lowrie (1988) and two from Channell et al. (1984) (Table 1).

TABLE 1

Location and age of northern Umbrian paleomagnetic sampling sites. Stratigraphic ages converted to absolute ages using Kent and Gradstein (1985)

ID	Section	Formation	Stratigraphic age	Age (Ma)	Reference
1	Bosso	Scaglia rossa (lower)	Santonian	87 ± 1	This paper
2	Bosso	Maiolica (upper)	Hauterivian–Barremian	125 ± 5	This paper
3	Bosso	Maiolica (lower)	Berriasian–Valanginian	138 ± 6	This paper
4	Bosso	Maiolica (lower)	Berriasian–Valanginian	138 ± 6	Hirt and Lowrie (1988)
5	Bosso	Calcari Diasparini (upper)	Kimmeridgian–Tithonian	150 ± 6	This paper
6	Bosso	Ammonitico rosso	Toarcian–Aalenian	188 ± 5	This paper
7	Bosso	Corniola	Pliensbachian	195 ± 3	This paper
8	Burano	Scaglia rossa (lower)	Late Santonian	85 ± 1	This paper
9	Burano	Maiolica (lower)	Berriasian–Valanginian	138 ± 6	This paper
10	Burano	Ammonitico rosso (W)	Toarcian–Aalenian	188 ± 5	This paper
11	Burano	Ammonitico rosso (E)	Toarcian–Aalenian	188 ± 5	This paper
12	Burano	Corniola	Pliensbachian	195 ± 3	This paper
13	Sentino	Scaglia rossa (lower)	Late Turonian	89 ± 1	This paper
14	Sentino	Maiolica (upper)	Hauterivian–Barremian	125 ± 5	This paper
15	Sentino	Maiolica (lower)	Berriasian–Valanginian	138 ± 6	This paper
16	Sentino	Maiolica (lower)	Berriasian–Valanginian	138 ± 6	Hirt and Lowrie (1988)
17	Sentino	Calcari Diasparini (upper)	Kimmeridgian–Tithonian	150 ± 6	Channell et al. (1984)
18	Sentino	Ammonitico rosso	Toarcian–Aalenian	188 ± 5	This paper
19	Sentino	Ammonitico rosso/ Monte Serrone	Toarcian–Aalenian	188 ± 5	Channell et al. (1984)
20	Sentino	Corniola	Pliensbachian	195 ± 3	This paper

The Corniola and Ammonitico Rosso sites are lithostratigraphically constrained to the Pliensbachian and Toarcian–Aalenian, respectively. The Maiolica spans the late Tithonian to Barremian interval in Umbria, however we located our sites either close to the top or close to the base of the sequence, allowing us to divide the sites into a Hauterivian–Barremian group and a Berriasian–Valanginian group (Table 1). Paleontological analysis of samples from the three Scaglia sites yielded foraminifera and nannofossils indicating a late Turonian–Santonian age (*Marginotruncana sigali* planktonic formaminiferal zone for Site 13, *Dicarinella asymetrica* foraminiferal zone for Site 1 and Site 8) (E. Erba and I. Premoli-Silva, pers. commun., 1992).

Paleomagnetic analysis

All samples were progressively thermally demagnetized at 50 or 25°C increments and the magnetization measured using a 2G-755R cryogenic magnetometer housed in the magnetically shielded laboratory at the University of Florida. The temperature range defining the characteristic (high unblocking temperature) magnetization component was determined by eye from orthogonal projections of demagnetization data (e.g., Figs. 2 and 3). The directions of the characteristic magnetization component were calculated using the standard principal component analysis (Kirschvink, 1980). Component directions within site were then combined using Fisher (1953) statistics (Table 2).

The sampled Scaglia is a white to pink pelagic limestone. Orthogonal projections of thermal demagnetization data (Fig. 2) indicate that the characteristic component is defined over a wide temperature range. The maximum unblocking temperature exceeds the Curie temperature for pure magnetite for some sites (e.g., Site 1, Fig. 2) but not for others (Site 13, Fig. 2), indicating that pigmentary hematite sometimes contributes to the characteristic magnetization component particularly in reddened samples.

The Maiolica is a white-gray cherty pelagic limestone. The natural remanent magnetization (NRM) intensities for the Maiolica limestones

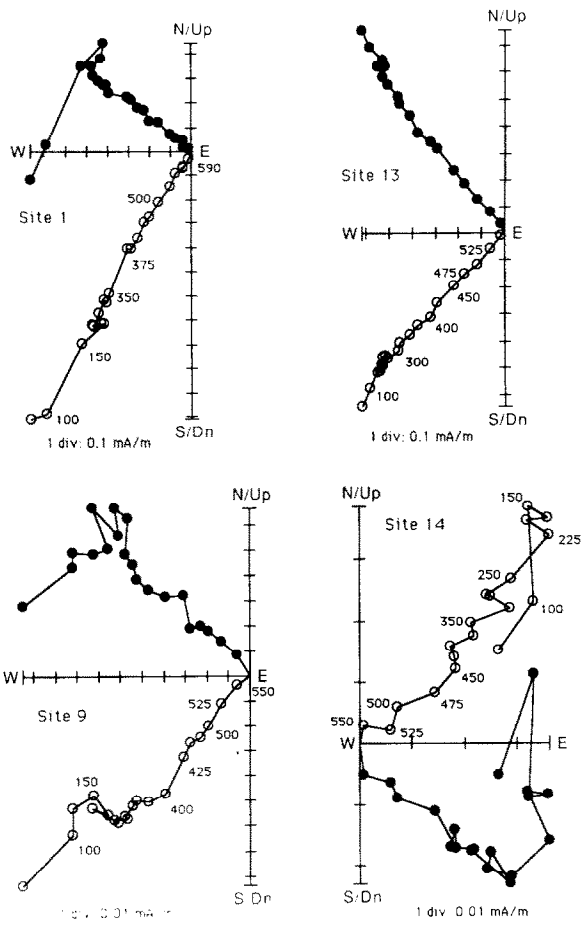


Fig. 2. Orthogonal projection of thermal demagnetization data for two samples of Scaglia (above) and two samples of Maiolica (below). Tilt corrected. Open and closed symbols represent projection on the vertical and horizontal planes, respectively. Temperatures given in °C.

(Fig. 2) are about an order of magnitude lower than those for the Scaglia. The more erratic decay of the magnetization vector during thermal demagnetization for the Maiolica (e.g., Site 14, Fig. 2) is a reflection of the weak magnetization intensities. The noise level of the 2G-755R cryogenic magnetometer is such that magnetization directions in samples with intensities of about 0.01 mA/m are not measured with optimal precision. Nonetheless, a characteristic magnetization component, with a maximum unblocking temperature close to 550°C, is usually well resolved (Fig. 2).

The Ammonitico Rosso is a pinkish to red nodular limestone. A characteristic magnetization component is clearly resolved over a large demag-

netization temperature range (Fig. 3). The maximum unblocking temperature does not generally exceed 570°C, implying that the hematite pigment is not an important contributor to the characteristic magnetization.

The Corniola is a white to pinkish pelagic limestone. Again, the characteristic magnetization is well defined over a large demagnetization temperature range from a few hundred degrees celsius to about 580°C (Fig. 3).

For ten of the twenty paleomagnetic sampling sites, the characteristic magnetization component has dual polarity within site (Table 2) and for each of these sites the polarities are stratigraphically constrained. For one site (Site 14), all characteristic components are reversed. For nine sites, all component directions within site have normal polarity. Of these nine sites, three are from Scaglia of Turonian–Santonian age which was deposited within the Late Cretaceous normal po-

TABLE 2

Site mean directions and age group mean directions for northern Umbrian sites

ID	Age		N	Pol.	k	α_{95}	In situ		Till corrected	
	Group	Ma					Dec.	Inc.	Dec.	Inc.
1	A	87 ± 1	18	N	268.7	2.1	339.4	0.7	308.0	46.8
2	B	125 ± 5	11	N	116.9	4.2	322.4	24.6	301.5	31.6
3	C	138 ± 6	10	N	32.6	8.6	316.1	29.5	298.5	37.7
4	C	138 ± 6	18	N/R	59.9	4.5	308.4	22.7	284.3	32.7
5	D	150 ± 6	24	N	45.3	4.4	310.4	27.8	293.5	34.3
6	E	191 ± 8	12	N/R	43.6	6.7	324.5	39.0	312.9	39.9
7	E	191 ± 8	11	N/R	111.6	4.3	333.0	50.4	314.7	56.4
8	A	85 ± 1	12	N	135.4	3.7	351.8	19.6	324.1	46.3
9	C	138 ± 6	8	N/R	41.0	8.8	258.1	12.0	292.4	35.0
10	E	191 ± 8	15	N/R	179.5	2.9	343.0	27.0	313.1	55.7
11	E	191 ± 8	15	N	87.5	4.1	293.0	40.8	320.3	47.7
12	E	191 ± 8	8	N	192.1	4.0	290.9	51.7	315.7	52.3
13	A	89 ± 1	18	N	304.2	2.0	333.9	42.8	313.9	36.9
14	B	125 ± 5	5	R	98.4	7.8	319.5	44.8	297.2	36.1
15	C	138 ± 6	10	N	125.1	4.3	311.4	39.9	294.3	32.6
16	C	138 ± 6	10	N/R	57.6	6.4	302.4	40.0	289.1	33.7
17	D	150 ± 6	163	N/R	9.2	3.9	310.9	39.2	292.7	31.3
18	E	191 ± 8	17	N/R	69.6	4.3	331.6	47.0	308.9	45.3
19	E	191 ± 8	93	N/R	9.6	5.0	330.1	42.2	303.3	46.5
20	E	191 ± 8	16	N/R	116.1	3.4	315.4	55.4	314.1	51.6

Means

Age Group	N	In situ Dec/Inc.	In situ k/α_{95}	Tilt corr. Dec/Inc.	Tilt corr. k/α_{95}	Pole Lat°N/Long°E	dp/dm
A	3	342.3/21.2	12.9/35.8	315.3/43.5	102.2/12.3	49.5/271.1	9.5/15.3
B	2	321.1/34.7		299.4/33.9		33.8/276.5	
C	5	298.7/30.7	11.3/23.8	291.6/34.4	274.6/ 4.6	28.4/282.4	3.0/ 5.3
D	2	310.6/33.5		293.1/32.8		28.8/280.4	
E	8	321.4/45.6	24.9/11.3	312.7/49.5	154.7/ 4.5	50.6/279.4	3.9/ 5.9
B + C + D	9	306.4/32.6	16.7/13.0	293.7/34.0	291.2/ 3.0	29.7/280.7	2.0/ 3.5

Poles after 20° clockwise rotation of group mean declinations:
Group A 62.96°N/248.79°E
Groups B + C + D 44.03°N/264.98°E
Group E 64.71°N/260.56°E

N: number of samples at site. Pol.: polarity of characteristic magnetization components. k, α_{95}, dp/dm are statistical parameters after Fisher (1953).

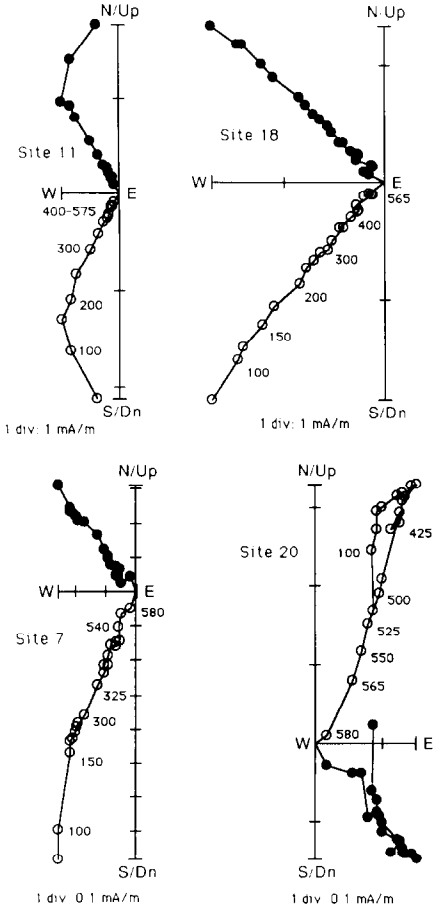


Fig. 3. Orthogonal projection of thermal demagnetization data for two samples of Ammonitico Rosso (above) and two samples of Corniola (below). Tilt corrected. Open and closed symbols represent projection on the vertical and horizontal planes, respectively. Temperatures given in °C.

larity superchron. The six remaining sites which exhibit only normal polarities are from the Maiolica (3), Ammonitico Rosso (1), Calcari Diasparini (1) and Corniola (1) (c.f. Tables 1 and 2).

The site mean *declinations* (Table 2) from the three river valleys (Bosso, Burano, Sentino), plotted as a function of age, show no clear relative rotation of the three sections (Fig. 4). When compared with the age group mean declinations from this area given by VandenBerg et al. (1978), the new mean declinations compare well for the Late Jurassic and Cretaceous, but are significantly different for the 191 ± 8 Ma (Pliensbachian–Aalenian) interval (Fig. 4; Table 3). The site mean declinations from Presale/Giordano/Gubbio are more westerly by about 20° relative to the Bosso/Burano/Sentino area (Fig. 4; Table 3), implying that Presale/Giordano/Gubbio have undergone anticlockwise rotation relative to the Bosso/Burano/Sentino region.

The Jurassic and Early Cretaceous site mean *inclinations* for the new dataset are consistent with those from Presale/Giordano/Gubbio but are significantly different from the age group mean inclinations of VandenBerg et al. (1978) (Fig. 5; Table 3). If the Jurassic and Early Cretaceous (pre-120 Ma) age group mean inclinations of VandenBerg et al. (1978) are plotted as site mean inclinations, the error bars overlap with those of some of the lower inclination data (Fig. 5). Nonetheless, the mean inclinations for Kimmeridgian–Tithonian and Berriasian–Valanginian data of VandenBerg et al. (1978) are anomalously high (Table 3). We attribute this to the inadequacy of alternating field demagnetization, used predominantly by VandenBerg et al. (1978), to resolve the primary single-component magnetization. High coercivity magnetization components

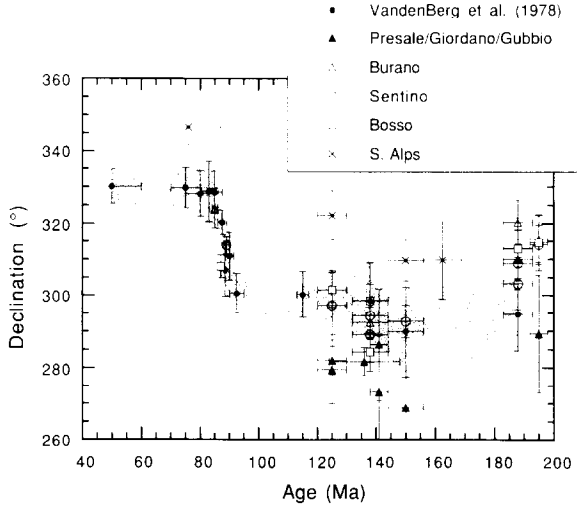


Fig. 4. Site mean declinations from three adjacent sections (Burano, Sentino, Bosso) compared to age group mean declinations of VandenBerg et al. (1978), and site mean declinations from Presale/Giordano/Gubbio (Lowrie and Alvarez, 1984; Cirilli et al., 1984; Márton and D'Andrea, 1992) and from the Southern Alps (Channell et al., 1992). Stratigraphic ages converted to absolute ages using Kent and Gradstein (1985). Error bar half-length for declination is $\alpha_{95}/\text{Cos I}$. The stippled region indicates the error bar limits for the data of VandenBerg et al. (1978). Plienbachian to Barremian (200–120 Ma) data are listed in Table 3.

of recent origin may account for the higher site mean inclinations in the VandenBerg et al. (1978) dataset.

A recent analysis of 46 paleomagnetic site mean directions from the Southern Alps (Belluno Basin, Trento Plateau, SE Lombardian Basin) has resulted in mean directions for the Late Cretaceous (76 ± 11 Ma), middle Cretaceous (125 ± 5 Ma), Kimmeridgian–Oxfordian (150 ± 6 Ma) and Callovian–Oxfordian (162 ± 7 Ma) (Channell et al., 1992). Comparison of the Southern Alpine mean declinations with the Umbrian data indicates that the Umbrian declinations are systematically lower by about 20° (Fig. 4; Table 3). This is interpreted to indicate 20° anticlockwise rotation of the Bosso/Burano/Sentino region, relative to

TABLE 3

Pliensbachian–Barremian site mean directions from northwestern Umbria. Southern Alpine mean inclinations have been adjusted to compensate for the latitudinal difference between Southern Alpine and Umbrian sampling regions

Location	Reference (site ID in this study)	Stratigraphic age	Tilt corrected		k	α_{95}
			Dec.	Inc.		
Presale	Márton and D'Andrea (1992)	Pliensbachian	289.3	54.0	21.0	9.6
Bosso	This paper (site 7)	Pliensbachian	314.7	56.4	111.6	4.3
Burano	This paper (site 12)	Pliensbachian	315.7	52.3	192.1	4.0
Sentino	This paper (site 20)	Pliensbachian	314.1	51.6	116.1	3.4
Presale	Márton and D'Andrea (1992)	Toarcian–Bajocian	310.0	50.8	57.0	5.3
Bosso	This paper (site 6)	Toarcian–Aalenian	312.9	39.9	43.6	6.7
Burano	This paper (site 10)	Toarcian–Aalenian	313.1	55.7	179.5	2.9
Burano	This paper (site 11)	Toarcian–Aalenian	320.3	47.7	87.5	4.1
Sentino	This paper (site 18)	Toarcian–Aalenian	308.9	45.3	69.6	4.3
Sentino	Channell et al. (1984) (site 19)	Toarcian–Aalenian	303.3	46.5	9.6	5.0
Sentino	VandenBerg et al. (1978)	Toarcian–Aalenian	290.2	51.2		9.3
Sentino	VandenBerg et al. (1978)	Toarcian–Aalenian	301.4	49.2		9.5
Southern Alps	Channell et al. (1992)	Callovian–Oxfordian	309.8	38.8	226.6	8.2
Presale	Márton and D'Andrea (1992)	Kimmeridgian–Tithonian	268.8	33.6	73.0	7.1
Bosso	This paper (site 5)	Kimmeridgian–Tithonian	293.5	34.3	45.3	4.4
Bosso	VandenBerg et al. (1978)	Kimmeridgian–Tithonian	289.3	48.4		16.2
Bosso	VandenBerg et al. (1978)	Kimmeridgian–Tithonian	290.4	51.3		9.6
Sentino	Channell et al. (1984) (site 17)	Kimmeridgian–Tithonian	292.7	31.3	9.2	3.9
Southern Alps	Channell et al. (1992)	Kimmeridgian–Tithonian	309.7	28.6	89.1	4.9
Fonte Giordano	Cirilli et al. (1984)	Tithonian–Hauterivian	281.7	33.5	835.7	3.2
Bosso	VandenBerg et al. (1978)	Early Neocomian	304.7	53.1		10.7
Bosso	VandenBerg et al. (1978)	Early Neocomian	293.7	41.9		7.7
Bosso	This paper (site 3)	Berriasian–Valanginian	298.5	37.7	32.6	8.6
Bosso	Hirt and Lowrie (1988) (site 4)	Berriasian–Valanginian	284.3	32.7	59.9	4.5
Burano	Cirilli et al. (1984)	Berriasian	286.3	27.8	20.4	13.7
Burano	This paper (site 9)	Berriasian–Valanginian	292.4	35.0	41.0	8.8
Sentino	This paper (site 15)	Berriasian–Valanginian	294.3	32.6	125.1	4.3
Sentino	Hirt and Lowrie (1988) (site 16)	Berriasian–Valanginian	289.1	33.7	57.6	6.4
Gubbio	Cirilli et al. (1984)	Berriasian	273.2	32.3	13.1	13.1
Presale	Márton and D'Andrea (1992)	Hauterivian–Barremian	279.5	40.0	31.0	7.3
Presale	Lowrie and Alvarez (1984)	Hauterivian–Barremian	282.0	41.0	18.3	2.9
Bosso	This paper (site 2)	Hauterivian–Barremian	301.5	31.6	116.9	4.2
Sentino	This paper (site 14)	Hauterivian–Barremian	297.2	36.1	98.4	7.8
Southern Alps	Channell et al. (1992)	Hauterivian–Barremian	322.2	37.5	87.3	5.2

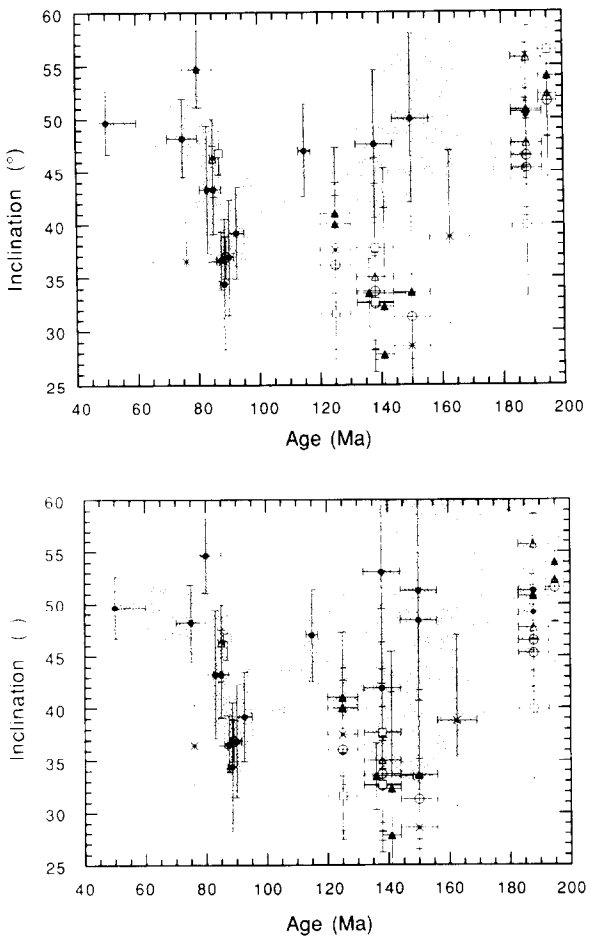


Fig. 5. Site mean inclinations from northwestern Umbria. Key as for Figure 4. Error bar half-length for inclination is α_{95}. The stippled region indicates (above) the error bar limits for age-group mean inclinations of VandenBerg et al. (1978), and (below) error bars limits for individual sites for the pre-120 Ma data of VandenBerg et al. (1978). Pliensbachian to Barremian (200–120 Ma) data are listed in Table 3. Southern Alpine inclinations have been adjusted to compensate for the latitudinal difference between Umbrian and Southern Alpine sampling sites.

the Southern Alps, during Neogene deformation of the Umbrian fold and thrust belt. The magnitude and sense of the rotation is comparable with that advocated for this region by VandenBerg et al. (1978), although these authors interpreted the rotation as a rotation of peninsular Italy. The site mean declinations from Presale (Márton and D'Andrea, 1992) imply about 40° anticlockwise rotation relative to the Southern Alps, suggesting that the magnitude of rotation increases northwestward from the Bosso/Burano/Sentino area.

The twenty new site mean directions from Umbria can be subdivided into five age groups (Table 2). The site mean directions for Group A (Turonian–Santonian) provide a positive fold test, significant at the 95% confidence level using either the criteria of McElhinny (1964) or McFadden (1990) (Fig. 6a). For age groups B, C and D (Kimmeridgian–Barremian), the site mean directions (Fig. 6b) are not significantly different from one another, and we therefore combine the nine site mean directions from these age groups and compute a single mean (Table 2). These site mean directions pass the fold test at the 99%

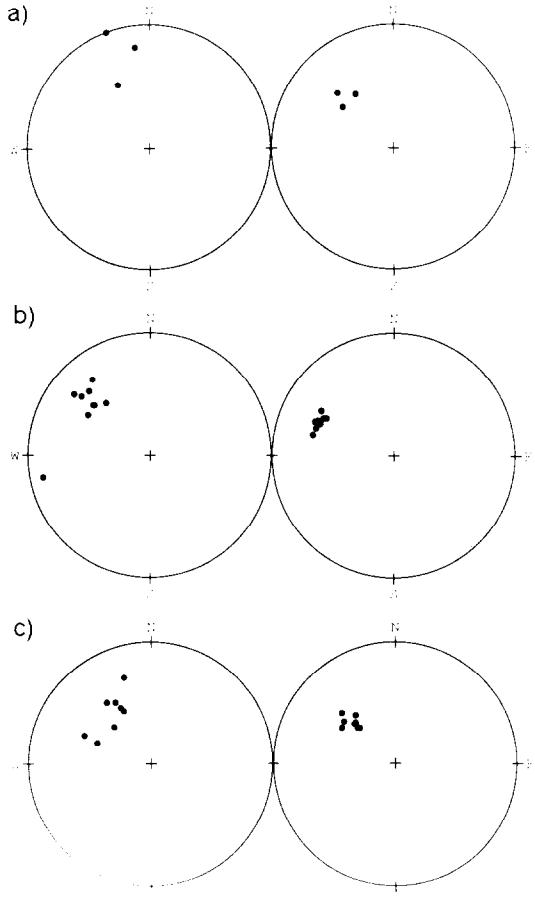


Fig. 6. Site mean directions from northern Umbria before (left) and after (right) structural tilt correction. Equal area projection. All inclinations are down (positive). (a) Age Group A (Late Cretaceous). (b) Age groups B–D (Late Jurassic and Early Cretaceous). (c) Age Group E (Early Jurassic).

confidence level (Fig. 6b) using either the McElhinny (1964) or McFadden (1990) criteria. The eight Pliensbachian–Aalenian sites constitute age group E. The site mean directions of this age group also pass the fold test at the 99% confidence level (Fig. 6c).

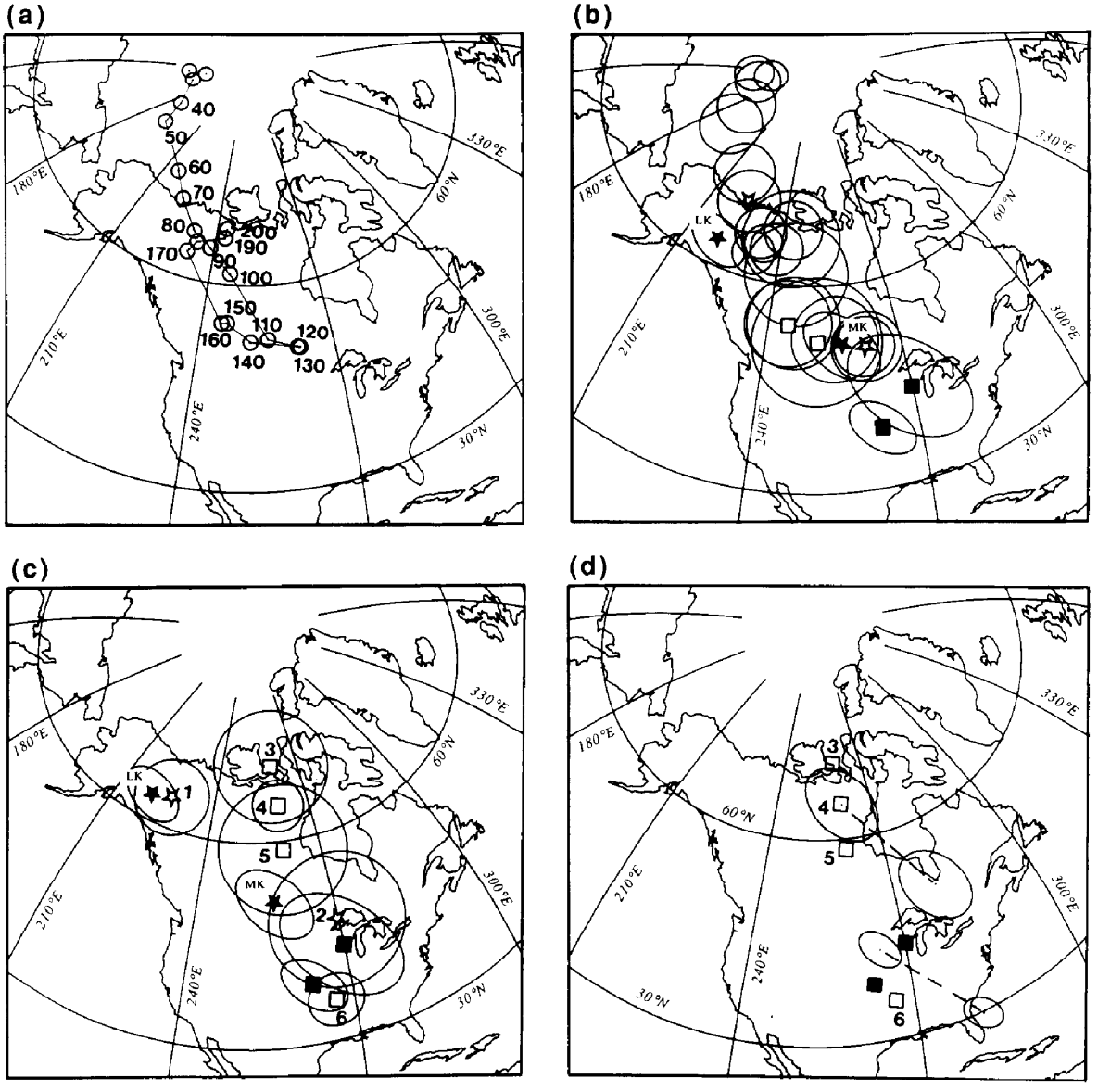


Fig. 7. (a) Synthetic African apparent polar wander path (APWP) of Besse and Courtillot (1991) with ages in Ma. (b) Synthetic African APWP of Besse and Courtillot (1991) with α_{95} confidence circles, and with Southern Alpine poles for the Late Cretaceous (*LK*) and middle Cretaceous (*MK*) (closed stars) and the Late Jurassic (closed squares). Coeval poles on the APWP are indicated by open symbols (after Channell et al., 1992). (c) Southern Alpine poles for the Late Cretaceous (*LK*) and middle Cretaceous (*MK*) (closed stars) compared with the Late Cretaceous (*1*) and Early Cretaceous (*2*) African poles (open stars) (Hargraves, 1989). Late Jurassic poles from the Southern Alps (closed squares) are compared with composite poles from Europe and Gondwana, rotated to African coordinates, for the Early Jurassic (Pliensbachian–Toarcian) (*3*), Middle Jurassic (Bajocian–Bathonian) (*4*), and Late Jurassic (Callovian–Kimmeridgian) (*5*); and with the Late Jurassic New York kimberlite pole (*6*), also rotated to African coordinates (Van Fossen and Kent, 1990, 1992). (d) Umbrian poles for the Early Jurassic and Late Jurassic–Early Cretaceous, before and after compensation for 20° anticlockwise tectonic rotation of the sampling sites. After compensation the poles move to the northwest. The poles are compared with the Jurassic poles shown in (c).

Paleomagnetic poles in African coordinates

The recently published apparent polar wander path (APWP) for Africa of Besse and Courtillot (1991) is shown in Figure 7a, and with 95% confidence circles in Figure 7b. For the southern Alpine Jurassic and Cretaceous mean directions given by Channell et al. (1992), the pole positions

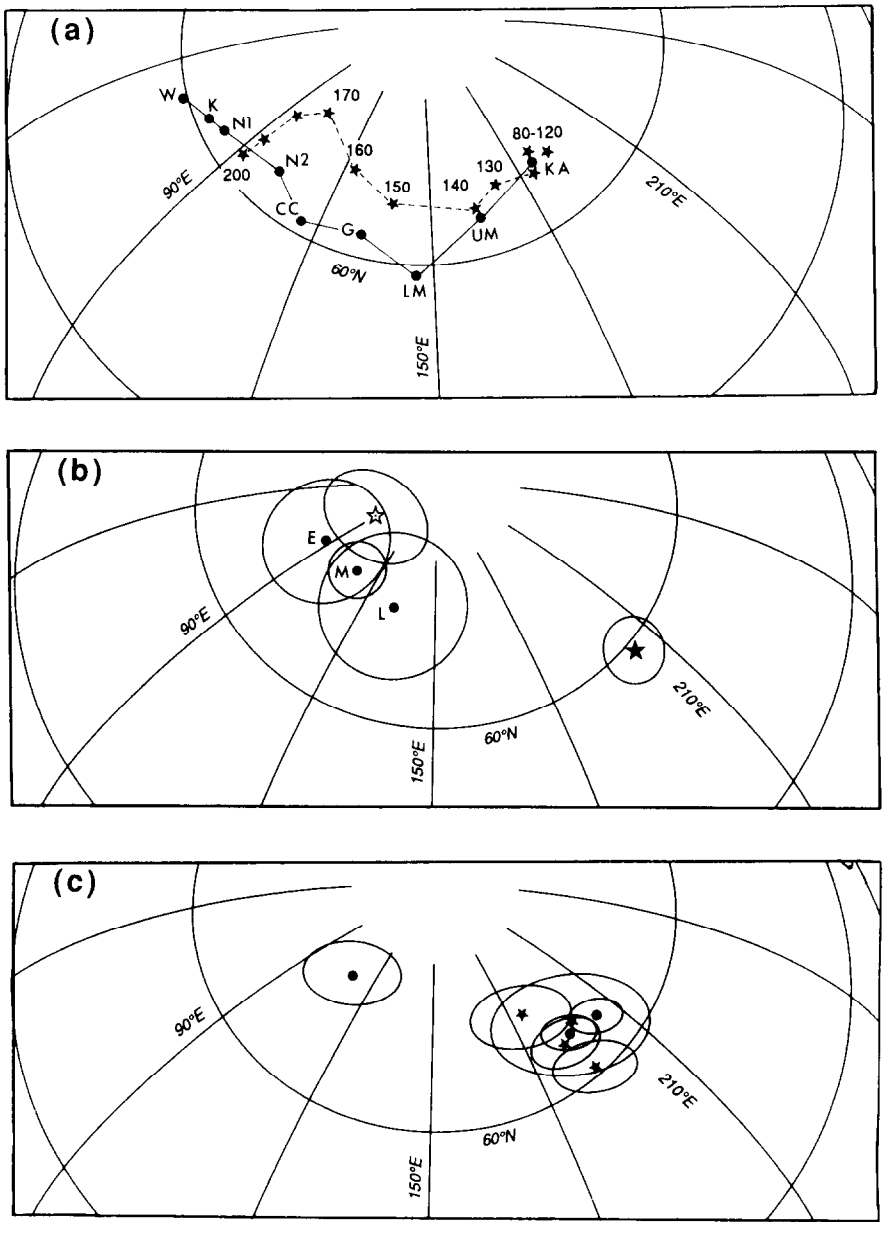


Fig. 8. (a) North American Mesozoic APWPs after Irving and Irving (1982) (stars) and May and Butler (1986) (dots). Ages in Ma are given for the Irving and Irving (1982) path. Key to May and Butler path: *W* = Wingate Fm. (Sinemurian); *K* = Kayenta Fm. (Pliensbachian); *N1* = Newark trend (195 Ma); *N2* = Newark Trend (179 Ma); *CC* = Coral Canyon (172 Ma); *G* = Glance Conglomerate (151 Ma); *LM* = Lower Morrison (early Tithonian); *UM* = Upper Morrison (145 Ma); *KA* = Cretaceous average (130–85 Ma). See May and Butler (1986) for references to individual poles. (b) North American coordinates. Open star: Moat Volcanics (166 Ma) pole (Van Fossen and Kent, 1990). Closed star: New York kimberlite pole (143 Ma) (Van Fossen and Kent, 1992). Eurasian and Gondwanan poles for Early Jurassic (Pliensbachian–Toarcian) (*E*), Middle Jurassic (Bajocian–Bathonian) (*M*), and Late Jurassic (Callovian–Kimmeridgian) (*L*), all rotated to North American coordinates (after Van Fossen and Kent, 1990). (c) Umbrian (dots) and Southern Alpine (stars) poles rotated to North American coordinates (see Table 4). The Pliensbachian–Aalenian Umbrian pole stands alone. The Late Jurassic and Cretaceous poles lie in a group.

conform closely with the African APWP for the Cretaceous, consistent with previous reviews of southern Alpine paleomagnetic data (Lowrie, 1986; Channell, 1986; Heller et al., 1989), but deviate from the African path for the Late Jurassic (Fig. 7b). The Southern Alpine Cretaceous poles lie close to African Cretaceous poles of Hargraves (1989) (Fig. 7c). The Late Jurassic Southern Alpine poles are not consistent with Gondwanan and European poles rotated to African coordinates, but lie close to the 143 Ma New York kimberlite pole of Van Fossen and Kent (1992), rotated to African coordinates (Fig. 7c). These data have been interpreted to indicate inaccuracy in the Besse and Courtillot (1991) African APWP for the Late Jurassic (Channell et al., 1992).

In order to test this conclusion in the light of the new paleomagnetic data from Umbria, the pole positions for the Kimmeridgian–Early Cretaceous (Groups B–D) and Pliensbachian–Aalenian (Group E) are plotted both before and after compensation for the 20° anticlockwise tectonic rotation (Fig. 7d). As expected, the rotated pole for the Kimmeridgian–Early Cretaceous coincides closely with the Late Jurassic poles from the Southern Alps (Fig. 7d). The rotated Pliensbachian–Aalenian pole is close to the Early Jurassic part of the Besse and Courtillot APWP (c.f. Figs. 7a and d), and to European and Gondwanan Jurassic poles rotated to African coordinates (Fig. 7d). The Late Cretaceous (Group A) rotated pole at 63°N 249°E lies close to the coeval (90 Ma) pole on the Besse and Courtillot APWP, but is based on only three sites, has a large confidence oval (Table 2) and is therefore not plotted in Figure 7d.

If the Southern Alpine and rotated Umbrian paleomagnetic data are a proxy for Mesozoic African paleomagnetic data, which appears likely in view of the general concordance of the data with the African APWP, we may conclude that Adria rotated with Africa during Mesozoic and Cenozoic times. The 20° anticlockwise rotation of northern Umbria relative to the African APWP (and to the Southern Alps) is a result of Neogene deformation of the sedimentary cover and does not signify Tertiary rotation of the Italian peninsula. If these conclusions are correct, the Southern Alpine and Umbrian paleomagnetic data indicate rapid apparent polar wander for Africa from high-latitude poles in Pliensbachian–Aalenian times to lower-latitude poles by Oxfordian times, requiring modification of the Besse and Courtillot African APWP.

Paleomagnetic poles in North American coordinates

There has been considerable recent debate concerning the North American Mesozoic APWP (see Van Fossen and Kent, 1990). The debate centers on the validity of the apparent polar wander path of May and Butler (1986) which differs from that of Irving and Irving (1982) (Fig. 8a). The notable difference between the paths is the high-latitude portion of the Irving and Irving (1982) polar wander path at about 170 Ma (Fig. 8a) which is the result of the inclusion in the Irving and Irving (1982) dataset of the White Mountain pole of Opdyke and Wensink (1966), which is not included in the May and Butler (1986) compilation. Van Fossen and Kent (1990) have restudied the White Mountains and produced a pole for the Moat Volcanics dated at 166 Ma. This is a high-latitude pole (open star in Fig. 8b) which essentially ratifies the high-latitude result of Opdyke and Wensink (1966). Hence, Van Fossen and Kent (1990) have argued for a high-latitude APWP in Early and Middle Jurassic times, similar to the Irving and Irving APWP. Van Fossen and Kent (1990) demonstrated that European and Gondwanan Jurassic poles rotated to North American coordinates (poles E, M and L in Fig. 8b) also give high-latitude poles, and have questioned the validity of key poles (such as CC, G and LM in Fig. 8a) in the May and Butler (1986) path. A recent result from kimberlites in New York dated at 143 Ma (Jurassic–Cretaceous boundary) implies a very different pole position by this time (closed star in Fig. 8b; Van Fossen and Kent, 1992).

To compare the Italian paleomagnetic data with the North American APWPs, the Southern Alpine and Umbrian poles are rotated to North American coordinates using the rotation poles of

Klitgord and Schouten (1986) for the opening of the central Atlantic between Africa and North America (Table 4). The Late Jurassic and Early Cretaceous rotated poles plot in a group (Fig. 8c) close to the result from the New York kimberlites (closed star in Fig. 8b) of Van Fossen and Kent (1992). However, the Pliensbachian–Aalenian rotated Umbrian pole (Fig. 8c) lies close to the high-latitude Jurassic poles in Figure 8b.

Existing data are consistent with rapid apparent polar wander in both North American and African coordinates in Middle Jurassic times. The Jurassic data from the Southern Alps and Umbria are pivotal to the timing of the apparent polar wander. The argument for high-latitude poles in North American and African coordinates is supported by the rotated Umbrian Pliensbachian–Aalenian (199–183 Ma) pole, the Moat Volcanics pole (~ 166 Ma, Callovian; Van Fossen and Kent, 1990) and by rotated Gondwanan and European poles for the Early (Pliensbachian–Toarcian), Middle (Bajocian–Bathonian) and Late (Callovian–Kimmeridgian) Jurassic (see Van Fossen and Kent, 1990). The lower-latitude pole position [in North American coordinates (Fig. 8c) and in African coordinates (Fig. 7c,d)] is supported by the Callovian–Oxfordian Southern Alpine pole, by the Kimmeridgian–Early Cretaceous poles from Umbria and the Southern Alps (Channell et al., 1992), by the 143 Ma New York kimberlite pole of Van Fossen and Kent (1992) and by the 145 Ma Swartruggens kimberlite African pole of Hargraves (1989).

Clearly, there is conflicting evidence as to when APWP for Africa and North America shifted to lower latitude. It should be mentioned that the Callovian–Oxfordian mean from the Southern Alps is based on only three sites (see Channell et al., 1992). However, the Kimmeridgian–Tithonian result from the Southern Alps is considerably more robust being based on eleven sites which pass the fold test and exhibit dual polarity within site (Channell et al., 1992). There remains the lingering doubt that the Late Jurassic sampling sites in Umbria and the Southern Alps may have been remagnetized in the Early Cretaceous. The deformation is largely Miocene and Pliocene in age in both sampling areas and, therefore, the positive fold tests do not tightly constrain the age of magnetization. The principal argument against remagnetization of the Kimmeridgian–Tithonian strata in the Southern Alps is the presence of stratigraphically constrained reversals at most sites, with well defined magnetic stratigraphies at four of the sites (see Channell and Grandesso, 1987). We tentatively conclude that the shift in

TABLE 4

Mesozoic paleomagnetic poles rotated to North American coordinates using rotation poles and angles of Srivastava and Tapscott (1986) for Eurasia–North America, and Klitgord and Schouten (1986) for Africa/Italy–North America

	Ref.	Age (Ma)	Pole		dp/dm (A_{95})	Rotn. Pole		Ω	Pole (N.Am.Coords.)	
			Lat°N	Long°E		Lat°N	Long°E		Lat°N	Long°E
NY Kimberlites (USA)	1	143	58.0	203.1	(3.8)				58.0	203.1
Moat Volcanics (USA)	2	166	81.6	89.7	5.1/7.1				81.6	89.7
Late Jurassic Mean (Eur)	2	~ 159	72.7	141.6	(8.9)	79.5	151.92	−25.59	73.9	131.4
Mid. Jurassic Mean (Gond)	2	~ 173							75.6	109.1
E. Jurassic Mean (Eur)	2	~ 192	71.1	104.2	(7.8)	79.5	151.92	−25.59	75.0	87.5
Late Cret. (S. Alps)	3	76 ± 11	63.4	220.2	2.7/4.5	81.35	−9.15	−22.87	65.4	191.7
Mid. Cret. (S. Alps)	3	125 ± 5	51.5	257.0	3.7/6.2	66.13	−19.0	−56.39	70.9	184.1
Kimm./Tith. (S. Alps)	3	150 ± 6	39.1	263.3	3.0/5.4	66.5	−18.1	−61.92	61.0	194.4
Call./Oxf. (S. Alps)	3	162 ± 7	44.0	269.8	6.0/10.0	67.15	−16.0	−64.70	66.6	196.8
Kimm.–Early Cret. (Umbria)	4	120–155	44.0	265.0	2.0/3.5	66.13	−19.0	−56.39	65.2	203.7
						66.5	−18.1	−61.92	66.1	194.3
Pliens./Aalen. (Umbria)	4	191 ± 8	64.7	260.6	3.9/5.9	66.95	−12.02	−75.55	75.3	108.4

References: (1) Van Fossen and Kent, 1992; (2) Van Fossen and Kent, 1990; (3) Channell et al. (1992); (4) This paper.

pole position to lower latitude (in both North American and African coordinates) occurred between Bathonian and Oxfordian times. This is consistent with most available data but implies error in the Late Jurassic (Callovian–Kimmeridgian) poles from Europe and in Late Jurassic part of the Besse and Courtillot (1991) APWP, and possibly underestimation of the age of the 166 Ma Moat Volcanics pole.

Acknowledgments

W. Lowrie and C. Klootwijk provided very helpful reviews of an earlier version of this paper. M. Van Fossen and D. Kent provided both a preprint of their paper on the New York kimberlite dykes and useful comments on an earlier draft of this paper. E. Erba and I. Premoli-Silva kindly determined the biostratigraphic age of the three Scaglia sites. Paleomagnetic work in Northern Italy has been supported by U.S. National Science Foundation (EAR 8804321 and OCE 8915697).

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